THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Real Numbers** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 12	Take the lowest powers of common primes
2	**(c)** 24	Take the highest powers of all primes
3	**(a)** 13	HCF of 65 and 117
4	**(c)** 27	LCM $+$ remainder
5	**(a)** non-terminating repeating	$210 = 2 \times 3 \times 5 \times 7$
6	**(d)** rational	$\sqrt{3} \times \sqrt{3} = 3$
7	**(d)** 15	Take the lowest powers of common primes
8	**(c)** 60	Take the highest powers of all primes
9	**(b)** 18	HCF of 72 and 126
10	**(c)** 62	LCM $+$ remainder

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Numbers exactly divisible: 240, 1024 (1); HCF $= 16$ (1).	1+1
12	LCM$(12, 16, 18) = 144$ (1); number $= 144 + 5 = 149$ (1).	1+1
13	$200 = 2^3 \times 5^2$ (1); places $=$ max of the powers of 2 and 5 $= 3$ (1).	1+1
14	Numbers exactly divisible: 391, 425 (1); HCF $= 17$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	HCF $\times$ LCM $=$ product of numbers (1); other $= \dfrac{9 \times 360}{72}$ (1) $= 45$ (1).	1+1+1
16	Required time $=$ LCM$(12, 15, 18)$ (1); $12 = 2^2 \times 3$, $15 = 3 \times 5$, $18 = 2 \times 3^2$ (1); LCM $= 180$ minutes (1).	1+1+1
17	Product $= 5^2 - 4 \times 3 = 13$ is rational (2); $5 + 2\sqrt{3}$ is irrational as sum of rational and irrational (1).	2+1
	OR: HCF $\times$ LCM $=$ product of numbers (1); other $= \dfrac{12 \times 504}{84}$ (1) $= 72$ (1).	1+1+1
18	Required time $=$ LCM$(6, 10, 14)$ (1); $6 = 2 \times 3$, $10 = 2 \times 5$, $14 = 2 \times 7$ (1); LCM $= 210$ minutes (1).	1+1+1
	OR: Product $= 7^2 - 9 \times 2 = 31$ is rational (2); $7 + 3\sqrt{2}$ is irrational as sum of rational and irrational (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Capacity $=$ HCF$(850, 680) = 170$ L (3); measures 5 and 4 (2).	3+2
	OR: Side $=$ HCF$(24, 16) = 8$ m (2); tiles $= \dfrac{24 \times 16}{64} = 6$ (3).	2+3
20	Capacity $=$ HCF$(612, 918) = 306$ L (3); measures 2 and 3 (2).	3+2
	OR: Side $=$ HCF$(45, 30) = 15$ m (2); tiles $= \dfrac{45 \times 30}{225} = 6$ (3).	2+3

Mistakes to avoid:

- Using HCF $\times$ LCM = product for **three** numbers; the result holds only for two numbers.
- Taking the *greatest* power for HCF or the *smallest* power for LCM; HCF uses smallest common powers, LCM uses greatest powers.
- Checking the denominator before reducing the fraction to lowest terms when testing for a terminating decimal.
- Assuming the product (or sum) of two irrational numbers is always irrational: $\sqrt{2} \times \sqrt{2} = 2$.
- In irrationality proofs, forgetting to state that a and b are co-prime at the start, which is what gives the contradiction.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Real Numbers** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** terminating	$312500 = 2^2 \times 5^7$
2	**(a)** rational	$\sqrt{5} \times \sqrt{5} = 5$
3	**(a)** 28	Take the lowest powers of common primes
4	**(c)** 36	Take the highest powers of all primes
5	**(c)** 21	HCF of 105 and 168
6	**(b)** 184	LCM + remainder
7	**(c)** terminating	$50 = 2 \times 5^2$
8	**(c)** rational	$\sqrt{7} \times \sqrt{7} = 7$
9	**(b)** 36	Take the lowest powers of common primes
10	**(d)** 60	Take the highest powers of all primes

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	LCM$(15, 20, 25) = 300$ (1); number $= 300 + 6 = 306$ (1).	1+1
12	$32 = 2^5$ (1); places $= \max$ of the powers of 2 and 5 $= 5$ (1).	1+1
13	Numbers exactly divisible: 456, 696 (1); HCF $= 24$ (1).	1+1
14	LCM$(14, 21, 35) = 210$ (1); number $= 210 + 3 = 213$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	HCF $\times$ LCM = product of numbers (1); other $= \dfrac{8 \times 240}{48}$ (1) $= 40$ (1).	1+1+1
16	Required time = LCM$(8, 20, 30)$ (1); $8 = 2^3$, $20 = 2^2 \times 5$, $30 = 2 \times 3 \times 5$ (1); LCM $= 120$ minutes (1).	1+1+1
17	Product $= 6^2 - 1 \times 5 = 31$ is rational (2); $6 + 1\sqrt{5}$ is irrational as sum of rational and irrational (1).	2+1
	OR: HCF $\times$ LCM = product of numbers (1); other $= \dfrac{15 \times 450}{90}$ (1) $= 75$ (1).	1+1+1
18	Required time = LCM$(9, 12, 21)$ (1); $9 = 3^2$, $12 = 2^2 \times 3$, $21 = 3 \times 7$ (1); LCM $= 252$ minutes (1).	1+1+1
	OR: Product $= 8^2 - 4 \times 7 = 36$ is rational (2); $8 + 2\sqrt{7}$ is irrational as sum of rational and irrational (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Capacity = HCF$(420, 546) = 42$ L (3); measures 10 and 13 (2).	3+2
	OR: Side = HCF$(72, 48) = 24$ m (2); tiles $= \dfrac{72 \times 48}{576} = 6$ (3).	2+3
20	Capacity = HCF$(360, 504) = 72$ L (3); measures 5 and 7 (2).	3+2
	OR: Side = HCF$(96, 60) = 12$ m (2); tiles $= \dfrac{96 \times 60}{144} = 40$ (3).	2+3

Mistakes to avoid:

- Using HCF $\times$ LCM = product for **three** numbers; the result holds only for two numbers.
- Taking the *greatest* power for HCF or the *smallest* power for LCM; HCF uses smallest common powers, LCM uses greatest powers.
- Checking the denominator before reducing the fraction to lowest terms when testing for a terminating decimal.
- Assuming the product (or sum) of two irrational numbers is always irrational: $\sqrt{2} \times \sqrt{2} = 2$.
- In irrationality proofs, forgetting to state that a and b are co-prime at the start, which is what gives the contradiction.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Real Numbers** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 42	HCF of 84 and 126
2	**(d)** 31	LCM + remainder
3	**(a)** non-terminating repeating	$75 = 3 \times 5^2$
4	**(a)** rational	$\sqrt{11} \times \sqrt{11} = 11$
5	**(b)** 21	Take the lowest powers of common primes
6	**(b)** 84	Take the highest powers of all primes
7	**(d)** 15	HCF of 90 and 165
8	**(b)** 125	LCM + remainder
9	**(c)** terminating	$160 = 2^5 \times 5$
10	**(b)** rational	$\sqrt{13} \times \sqrt{13} = 13$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$625 = 5^4$ (1); places $= \max$ of the powers of 2 and 5 $= 4$ (1).	1+1
12	Numbers exactly divisible: 1250, 9375 (1); HCF $= 625$ (1).	1+1
13	LCM$(8, 18, 24) = 72$ (1); number $= 72 + 7 = 79$ (1).	1+1
14	$500 = 2^2 \times 5^3$ (1); places $= \max$ of the powers of 2 and 5 $= 3$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	HCF × LCM = product of numbers (1); other $= \dfrac{6 \times 396}{66}$ (1) $= 36$ (1).	1+1+1
16	Required time = LCM$(10, 16, 24)$ (1); $10 = 2 \times 5$, $16 = 2^4$, $24 = 2^3 \times 3$ (1); LCM $= 240$ minutes (1).	1+1+1
17	Product $= 9^2 - 9 \times 5 = 36$ is rational (2); $9 + 3\sqrt{5}$ is irrational as sum of rational and irrational (1).	2+1
	OR: HCF × LCM = product of numbers (1); other $= \dfrac{14 \times 420}{70}$ (1) $= 84$ (1).	1+1+1
18	Required time = LCM$(14, 18, 21)$ (1); $14 = 2 \times 7$, $18 = 2 \times 3^2$, $21 = 3 \times 7$ (1); LCM $= 126$ minutes (1).	1+1+1
	OR: Product $= 4^2 - 1 \times 11 = 5$ is rational (2); $4 + 1\sqrt{11}$ is irrational as sum of rational and irrational (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Capacity = HCF$(468, 624) = 156$ L (3); measures 3 and 4 (2).	3+2
	OR: Side = HCF$(84, 56) = 28$ m (2); tiles $= \dfrac{84 \times 56}{784} = 6$ (3).	2+3
20	Capacity = HCF$(720, 960) = 240$ L (3); measures 3 and 4 (2).	3+2
	OR: Side = HCF$(90, 54) = 18$ m (2); tiles $= \dfrac{90 \times 54}{324} = 15$ (3).	2+3

Mistakes to avoid:

- Using HCF × LCM = product for **three** numbers; the result holds only for two numbers.
- Taking the *greatest* power for HCF or the *smallest* power for LCM; HCF uses smallest common powers, LCM uses greatest powers.
- Checking the denominator before reducing the fraction to lowest terms when testing for a terminating decimal.
- Assuming the product (or sum) of two irrational numbers is always irrational: $\sqrt{2} \times \sqrt{2} = 2$.
- In irrationality proofs, forgetting to state that a and b are co-prime at the start, which is what gives the contradiction.

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